

Murtaza S. Ahmed

Undergrad Engineer | Distributed Systems, ML Systems & Infra (Distributed focus)

murtazaofficial@protonmail.com | +91 9390341984 | onerealti.github.io/portfolio | github.com/onerealti | www.linkedin.com/in/murahmed

Focusing on AI infrastructure, particularly GPU cluster orchestration, inference systems, model routing, resource allocation and workload scheduling.

Professional Experience

Undergraduate Research Assistant (Edge AI Systems) Department of Mechanical Engineering (MJCET)	Dec 2025 -- Present <i>Hyderabad, TG</i>
- Selected for an institution-funded R&D program under faculty supervision. - Built a complete edge AI system on NVIDIA Jetson Nano 4GB Developer Kit, integrating an IMX719 camera and soil sensors for real-time field operation. - Developed the on-device inference pipeline and executed real-time inference under compute and memory constraints. - Achieved 40–70 ms latency (14–20 FPS) using TensorRT (FP16) optimization. - Processed camera and sensor data continuously at 5–10 Hz with low-latency performance. - Designed a stable execution loop enabling reliable operation over 4–6 hour field deployments.	
Undergraduate Researcher (R&D Program) Department of Mechanical Engineering (MJCET)	March 2026 -- Present <i>Hyderabad, TG</i>
- Design and implementation of an Adaptive Multi-Objective Policy Engine for Resilient Orchestration in Heterogeneous LLM Clusters. - Researching adaptive workload scheduling and hardware-aware resource allocation strategies for heterogeneous LLM inference systems. - Exploring fault-tolerant orchestration approaches for distributed AI workloads across mixed compute environments.	
Builder & Systems Architect Verity (Decentralized Verification Platform)	July 2025 -- Present <i>Hyderabad, TG</i>
- Designing and implementing a system to verify digital media provenance using blockchain and decentralized storage. - Integrating AI modules for manipulation detection and automated claim extraction. - Developing media provenance infrastructure using distributed storage systems including IPFS and Arweave. - Building concurrent ingestion pipelines and immutable audit logging mechanisms for ML-based verification workflows.	

Projects

Smart Agri Four-Legged Bot (SAFL-B) <i>(C, Python, TensorRT, NVIDIA Jetson Nano, Arduino Due, SolidWorks, FEA, UART/Serial)</i>	May 2026
- Awarded 1st Place at the city-wide MAKEFORHYDERABAD Makeathon (organized by Titan & SIT Hyderabad) for autonomous robotics integration. - Designed a hybrid 3mm steel chassis and custom 3D-printed modular articulated planar legs featuring passive compliance and internal wire routing channels. - Validated structural load paths using Finite Element Analysis (FEA) under static vertical loads, bumps, and probe insertions with a minimum safety factor of 2.04. - Integrated a local CNN weed classification model on NVIDIA Jetson Nano, achieving real-time inference (9.2 FPS) using TensorRT FP16 acceleration. - Designed low-level kinematic control systems using an Arduino Due to orchestrate H-Bridge DC motor actuators and high-current ESCs.	

Technical Skills

Languages:	C, Python, Bash, PostgreSQL, Git
Systems:	Linux, Docker, Kubernetes
ML/Edge Computing:	TensorRT, NVIDIA Jetson Nano
Hardware & Interfaces:	UART/Serial, I2C, GPIO

Education

Bachelor of Engineering in Mechanical Engineering Muffakham Jah College of Engineering & Technology (Osmania University)	May 2028 <i>Hyderabad, TG</i>
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- Relevant Coursework: Scientific Computing, Embedded Systems Programming, System Modeling & Simulation